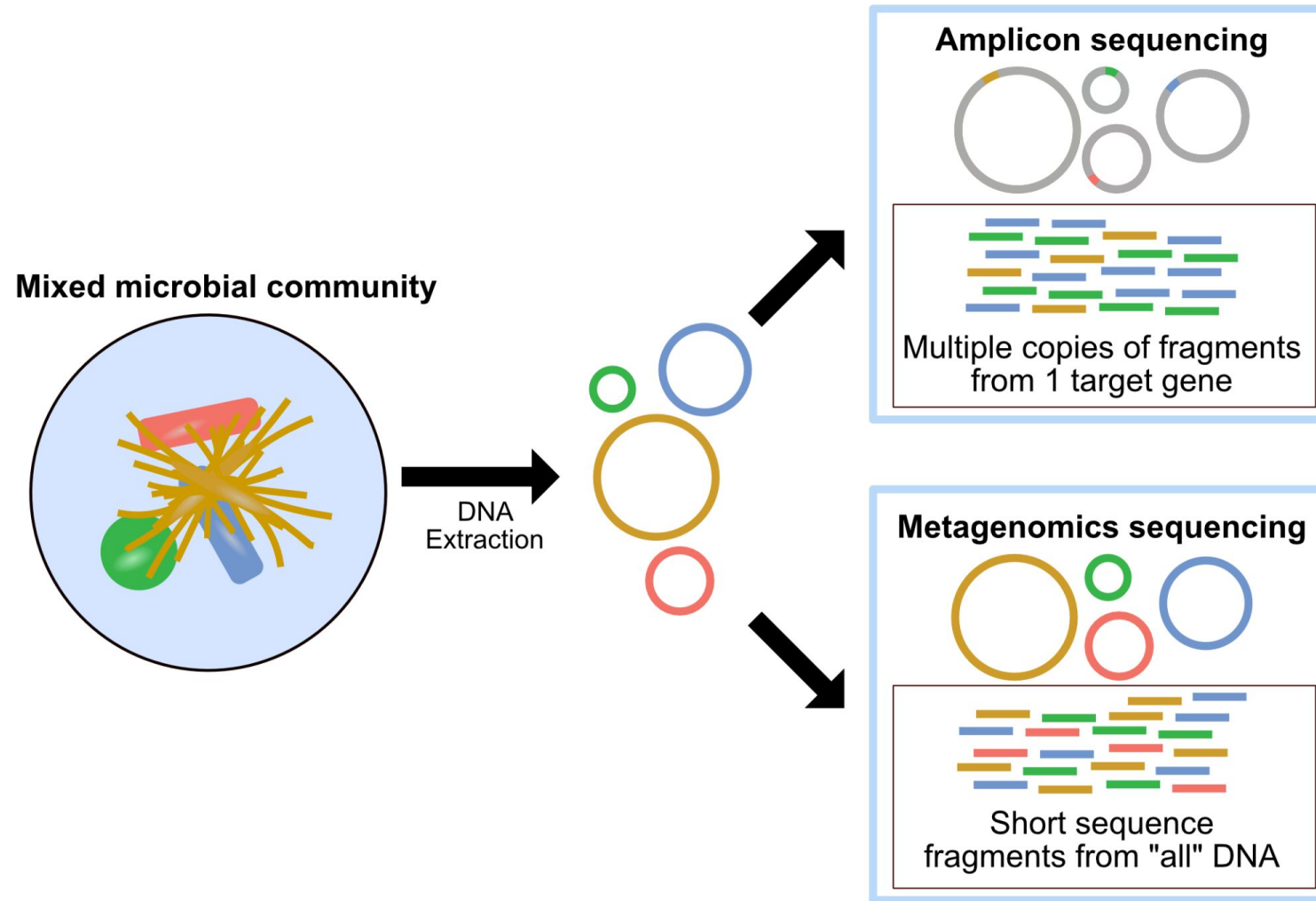
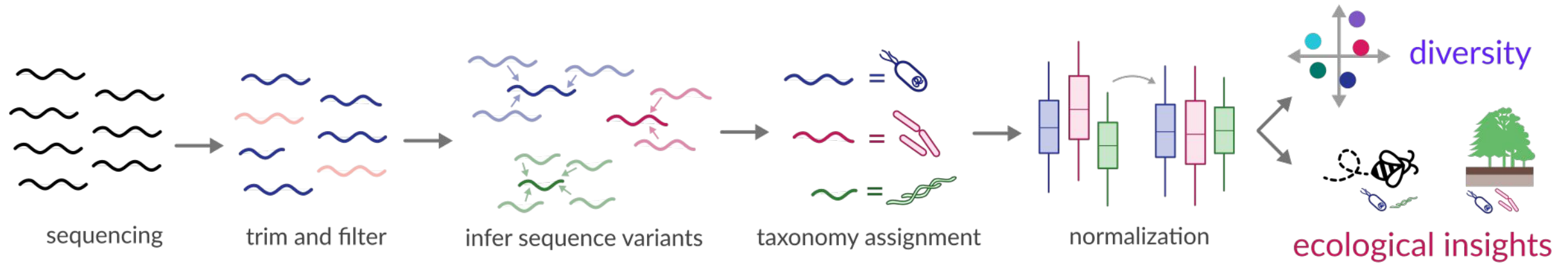


Microbial amplicon sequencing data analysis

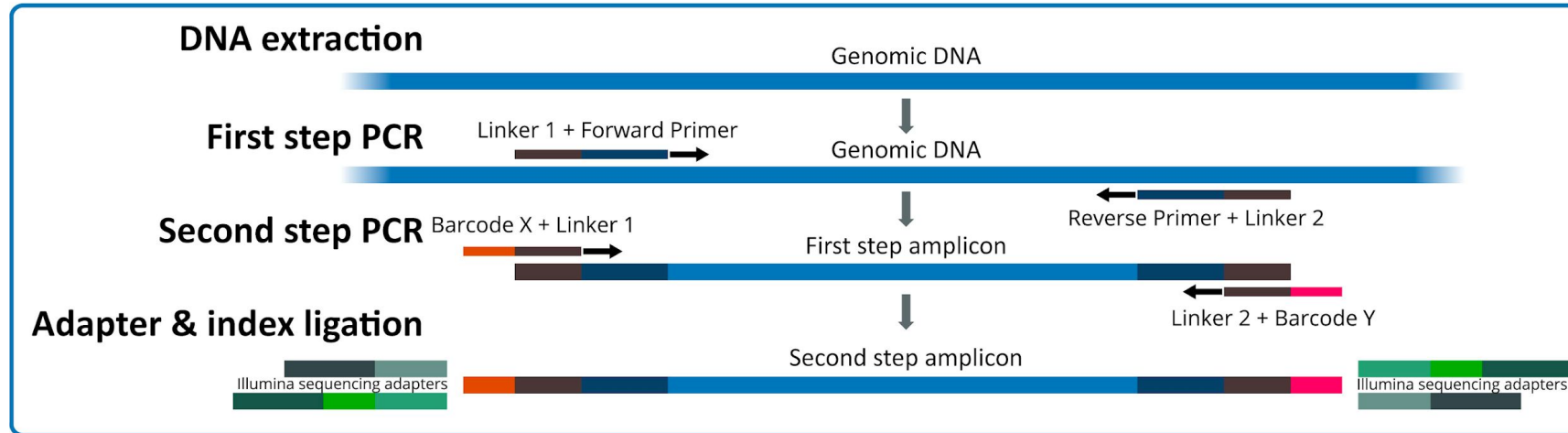
Amplicon sequencing of microbial community



Amplicon sequencing of microbial community



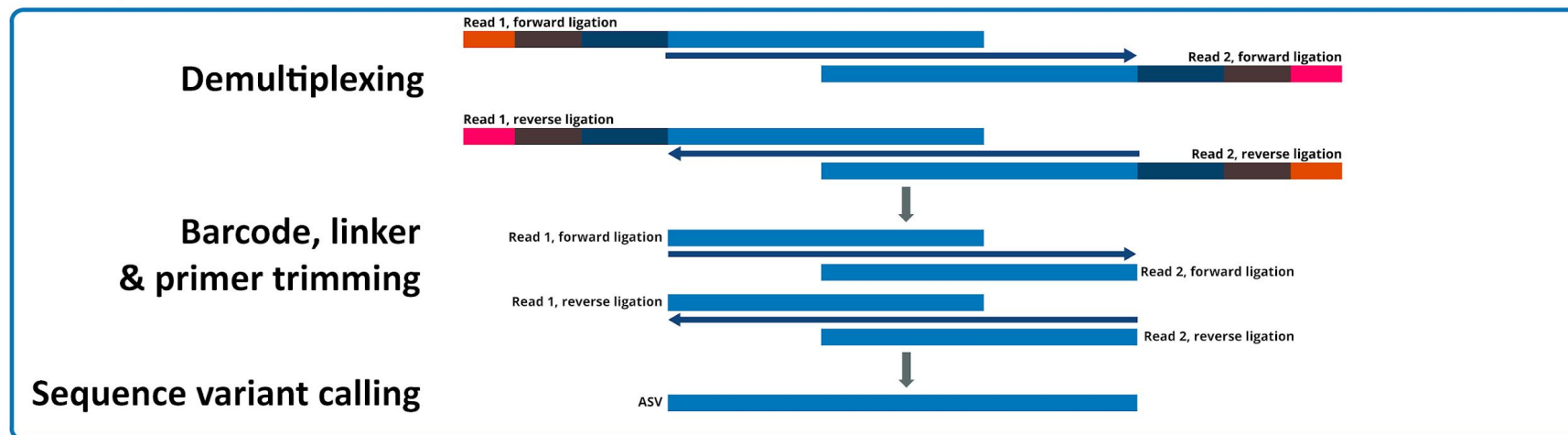
1. Amplicon generation



2. Sequencing

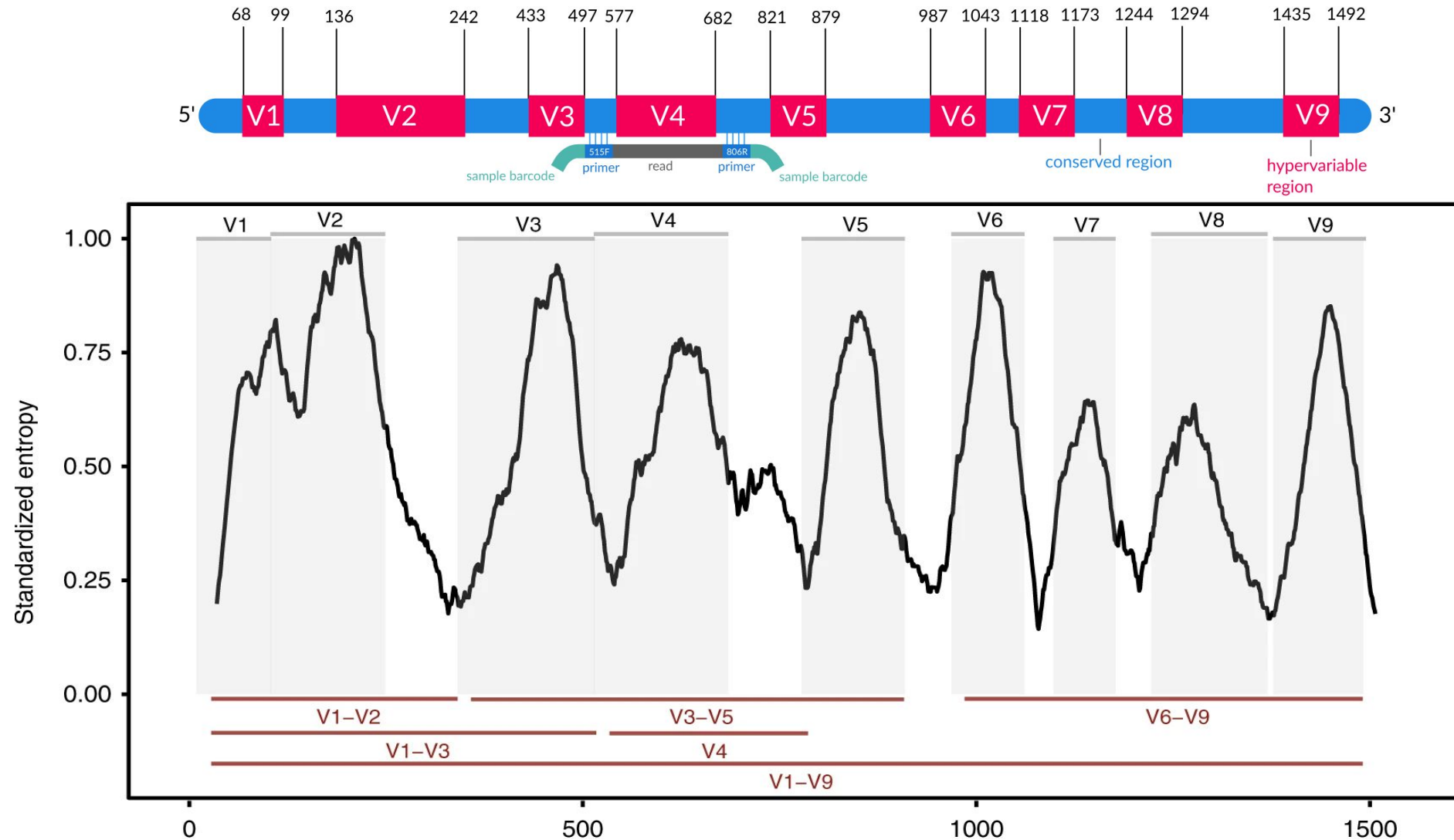


3. Amplicon sequence analysis



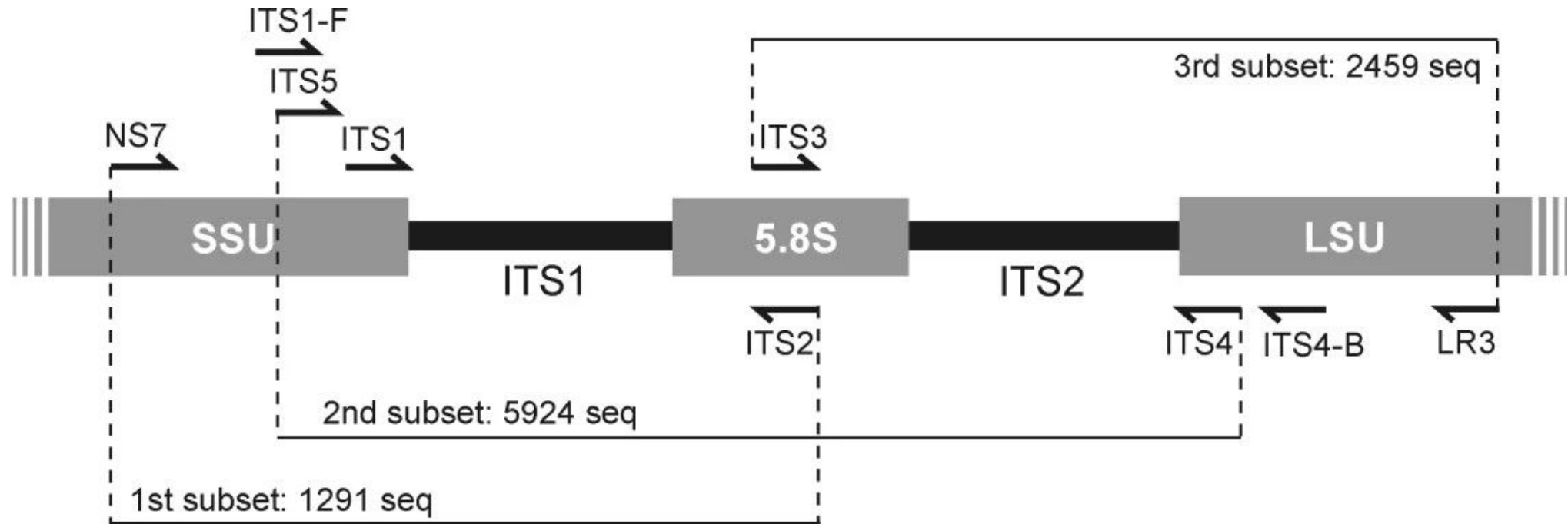
- Bacterial 16S rRNA gene
- Fungal nuclear ribosomal internal transcribed spacer (ITS)
- Eukaryote 18S rRNA gene
- Other single copy marker gene

Bacterial and archaeal 16S rRNA genes



Shannon entropy across the 16S rRNA gene based on the alignment of a single representative sequence for each known species present in the Greengenes database.

Fungal ITS and commonly used primers



Relative position of the primers, design of the subsets and number of sequences in each subset.

Let's decode

R10_1.fastq.gz

@M00758:1471:L5VVJ:1:1102:13606:2053 1:N:0:GGGCTTCAGGCC

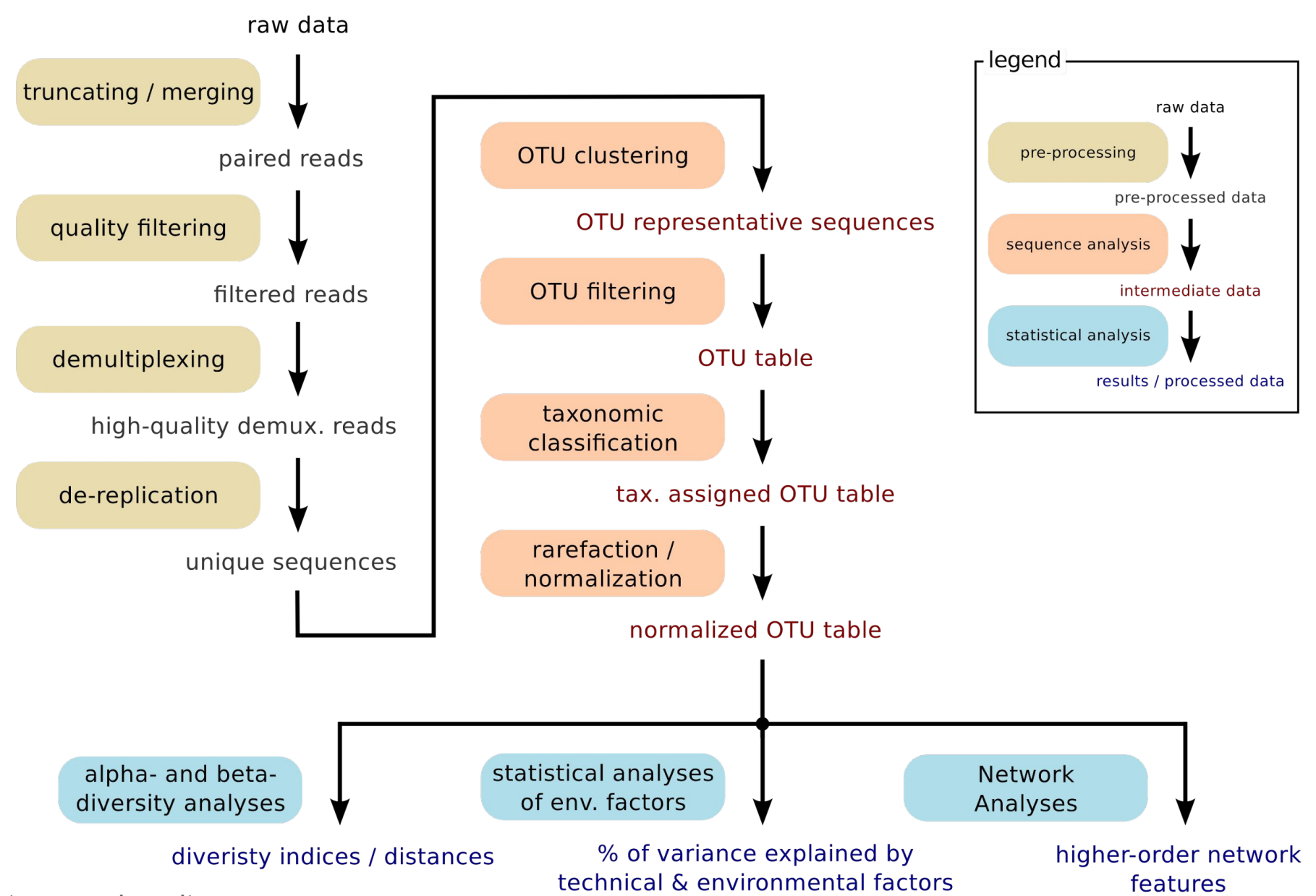
M00758	the unique instrument name
1471	the run id
L5VVJ	the flowcell id
1	flowcell lane
1102	tile number within the flowcell lane
13606	'x'-coordinate of the cluster within the tile
2053	'y'-coordinate of the cluster within the tile
1	the member of a pair, 1 or 2 (paired-end or mate-pair reads only)
N	Y if the read is filtered (did not pass), N otherwise
0	0 when none of the control bits are on, otherwise it is an even number
GGGCTTCAGGCC	index sequence

How computer talks to biology

ASCII TABLE

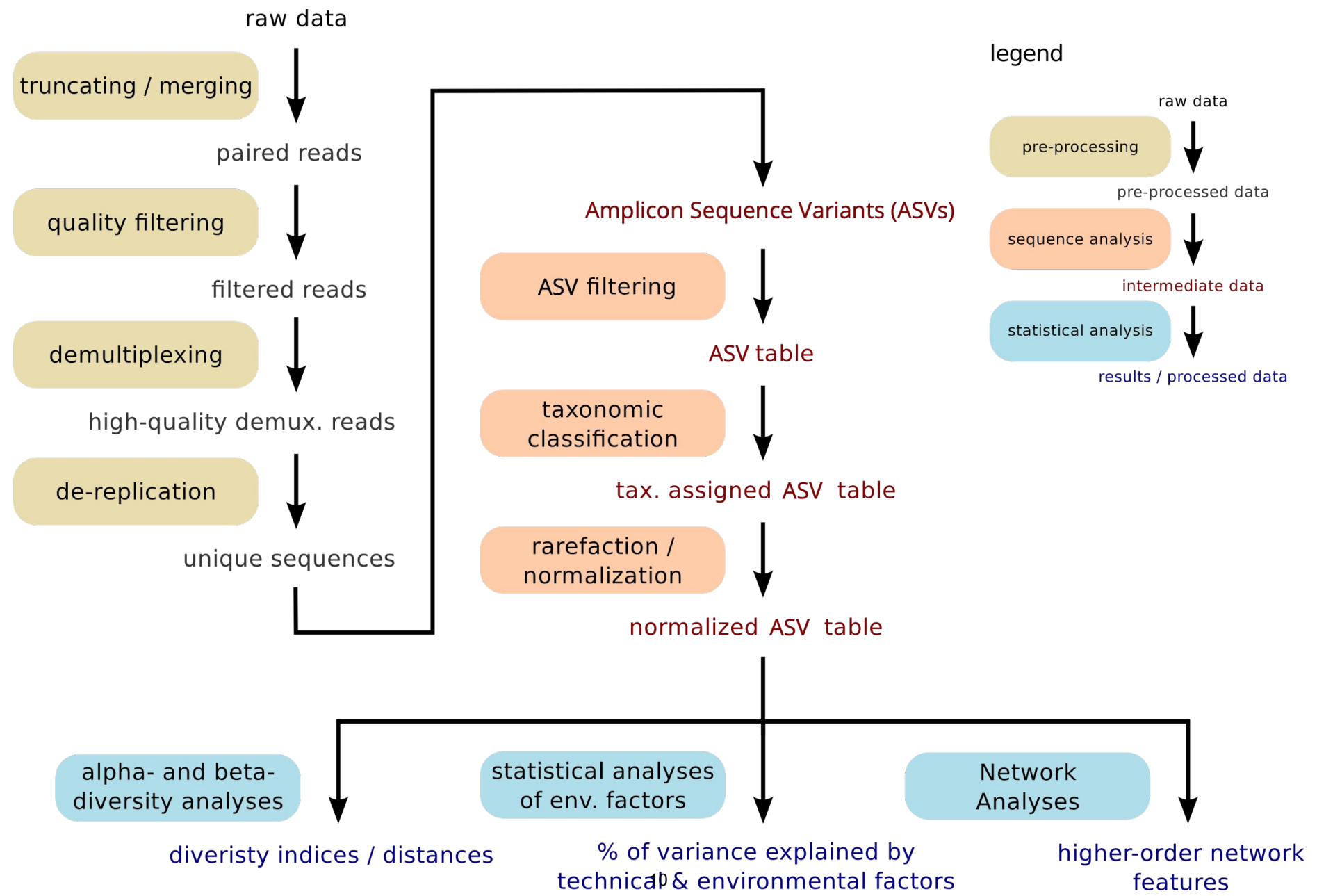
Decimal	Hexadecimal	Binary	Octal	Char	Decimal	Hexadecimal	Binary	Octal	Char	Decimal	Hexadecimal	Binary	Octal	Char
0	0	0	0	[NULL]	48	30	110000	60	0	96	60	1100000	140	`
1	1	1	1	[START OF HEADING]	49	31	110001	61	1	97	61	1100001	141	a
2	2	10	2	[START OF TEXT]	50	32	110010	62	2	98	62	1100010	142	b
3	3	11	3	[END OF TEXT]	51	33	110011	63	3	99	63	1100011	143	c
4	4	100	4	[END OF TRANSMISSION]	52	34	110100	64	4	100	64	1100100	144	d
5	5	101	5	[ENQUIRY]	53	35	110101	65	5	101	65	1100101	145	e
6	6	110	6	[ACKNOWLEDGE]	54	36	110110	66	6	102	66	1100110	146	f
7	7	111	7	[BELL]	55	37	110111	67	7	103	67	1100111	147	g
8	8	1000	10	[BACKSPACE]	56	38	111000	70	8	104	68	1101000	150	h
9	9	1001	11	[HORIZONTAL TAB]	57	39	111001	71	9	105	69	1101001	151	i
10	A	1010	12	[LINE FEED]	58	3A	111010	72	:	106	6A	1101010	152	j
11	B	1011	13	[VERTICAL TAB]	59	3B	111011	73	;	107	6B	1101011	153	k
12	C	1100	14	[FORM FEED]	60	3C	111100	74	<	108	6C	1101100	154	l
13	D	1101	15	[CARRIAGE RETURN]	61	3D	111101	75	=	109	6D	1101101	155	m
14	E	1110	16	[SHIFT OUT]	62	3E	111110	76	>	110	6E	1101110	156	n
15	F	1111	17	[SHIFT IN]	63	3F	111111	77	?	111	6F	1101111	157	o
16	10	10000	20	[DATA LINK ESCAPE]	64	40	1000000	100	@	112	70	1110000	160	p
17	11	10001	21	[DEVICE CONTROL 1]	65	41	1000001	101	A	113	71	1110001	161	q
18	12	10010	22	[DEVICE CONTROL 2]	66	42	1000010	102	B	114	72	1110010	162	r
19	13	10011	23	[DEVICE CONTROL 3]	67	43	1000011	103	C	115	73	1110011	163	s
20	14	10100	24	[DEVICE CONTROL 4]	68	44	1000100	104	D	116	74	1110100	164	t
21	15	10101	25	[NEGATIVE ACKNOWLEDGE]	69	45	1000101	105	E	117	75	1110101	165	u
22	16	10110	26	[SYNCHRONOUS IDLE]	70	46	1000110	106	F	118	76	1110110	166	v
23	17	10111	27	[ENG OF TRANS. BLOCK]	71	47	1000111	107	G	119	77	1110111	167	w
24	18	11000	30	[CANCEL]	72	48	1001000	110	H	120	78	1111000	170	x
25	19	11001	31	[END OF MEDIUM]	73	49	1001001	111	I	121	79	1111001	171	y
26	1A	11010	32	[SUBSTITUTE]	74	4A	1001010	112	J	122	7A	1111010	172	z
27	1B	11011	33	[ESCAPE]	75	4B	1001011	113	K	123	7B	1111011	173	{
28	1C	11100	34	[FILE SEPARATOR]	76	4C	1001100	114	L	124	7C	1111100	174	
29	1D	11101	35	[GROUP SEPARATOR]	77	4D	1001101	115	M	125	7D	1111101	175	}
30	1E	11110	36	[RECORD SEPARATOR]	78	4E	1001110	116	N	126	7E	1111110	176	~
31	1F	11111	37	[UNIT SEPARATOR]	79	4F	1001111	117	O	127	7F	1111111	177	[DEL]
32	20	100000	40	[SPACE]	80	50	1010000	120	P					
33	21	100001	41	!	81	51	1010001	121	Q					
34	22	100010	42	"	82	52	1010010	122	R					
35	23	100011	43	#	83	53	1010011	123	S					
36	24	100100	44	\$	84	54	1010100	124	T					
37	25	100101	45	%	85	55	1010101	125	U					
38	26	100110	46	&	86	56	1010110	126	V					
39	27	100111	47	'	87	57	1010111	127	W					
40	28	101000	50	(	88	58	1011000	130	X					
41	29	101001	51	)	89	59	1011001	131	Y					
42	2A	101010	52	*	90	5A	1011010	132	Z					
43	2B	101011	53	+	91	5B	1011011	133	[					
44	2C	101100	54	,	92	5C	1011100	134	\					
45	2D	101101	55	-	93	5D	1011101	135	]					
46	2E	101110	56	.	94	5E	1011110	136	^					
47	2F	101111	57	/	95	5F	1011111	137	_					

Amplicon sequencing data processing of natural communities at the OTU level



OTU: Operational taxonomic unit

Amplicon sequencing data processing of natural communities at the ASV level



Most commonly used pipeline

- QIIME2



https://gibbons-lab.github.io/isb_course_2024/16S/

- DADA2 (R based, available as plug-in in QIIME2)

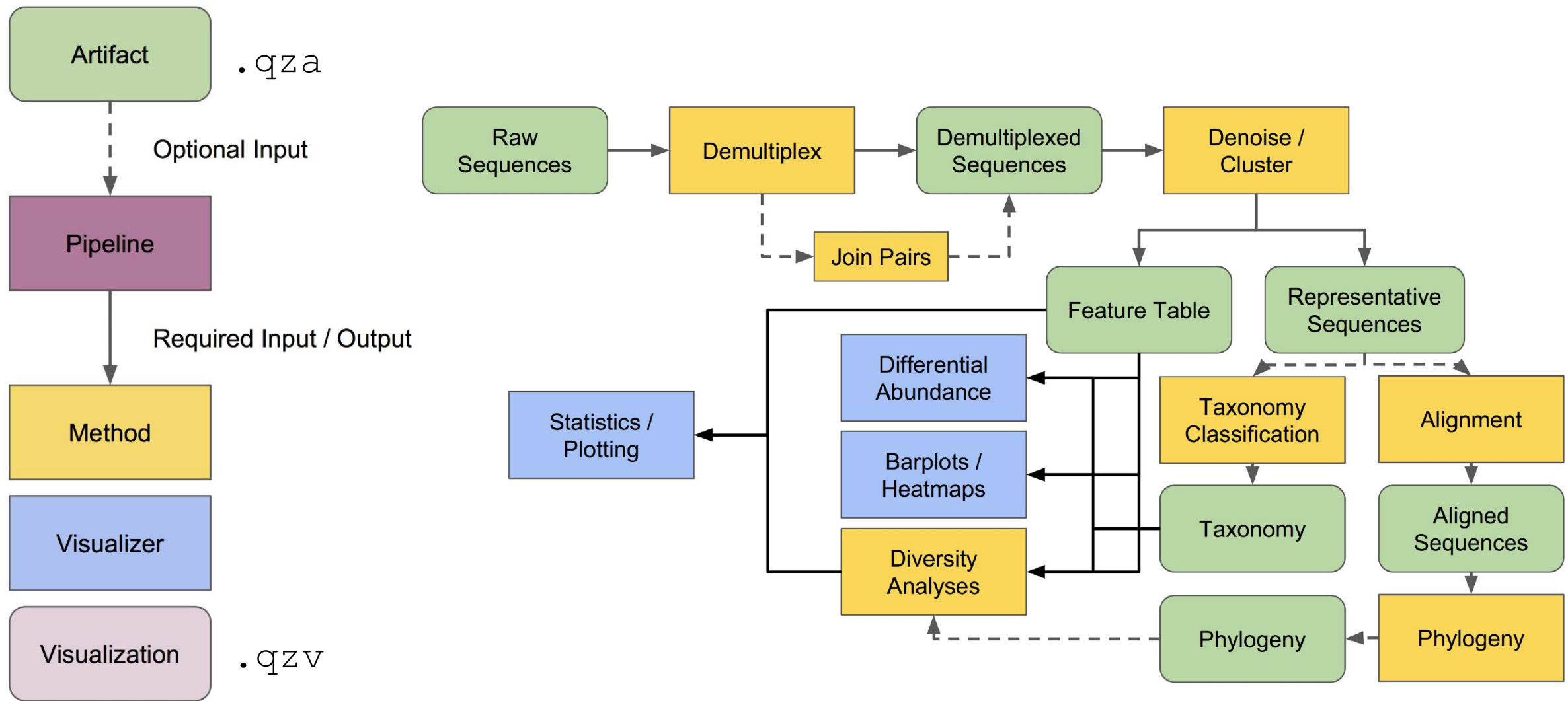


<https://benjjneb.github.io/dada2/index.html>

Other toolkits:

- [LotuS2](#) (developed by Hildebrand lab @Quadram & Earlham Institute)
- STARA (developed by Sina)

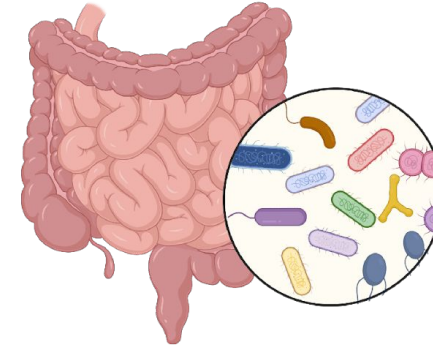
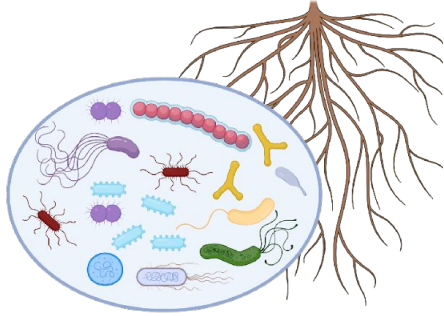
Conceptual overview of QIIME2



Processing of amplicon sequencing data using DADA2

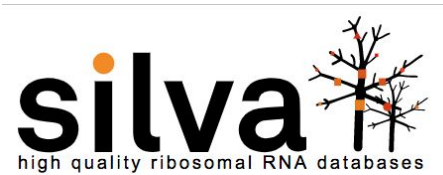
- Prepare the environment - R / Studio
- Data import and sample organization
- Quality control and filtering
- Error correction and read merging
- Creating and cleaning the ASV table
- Post-pipeline quality control
- Assign taxonomy
- Visualizing results with `Phyloseq`

Reference databases for natural and synthetic community profiling

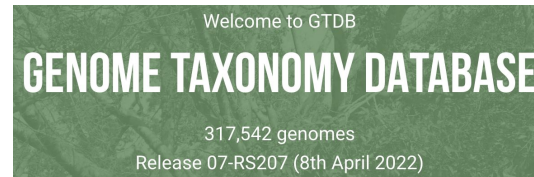


GREENGENES

<https://greengenes.secondgenome.com/>



<https://www.arb-silva.de/>



<http://rdp.cme.msu.edu/>



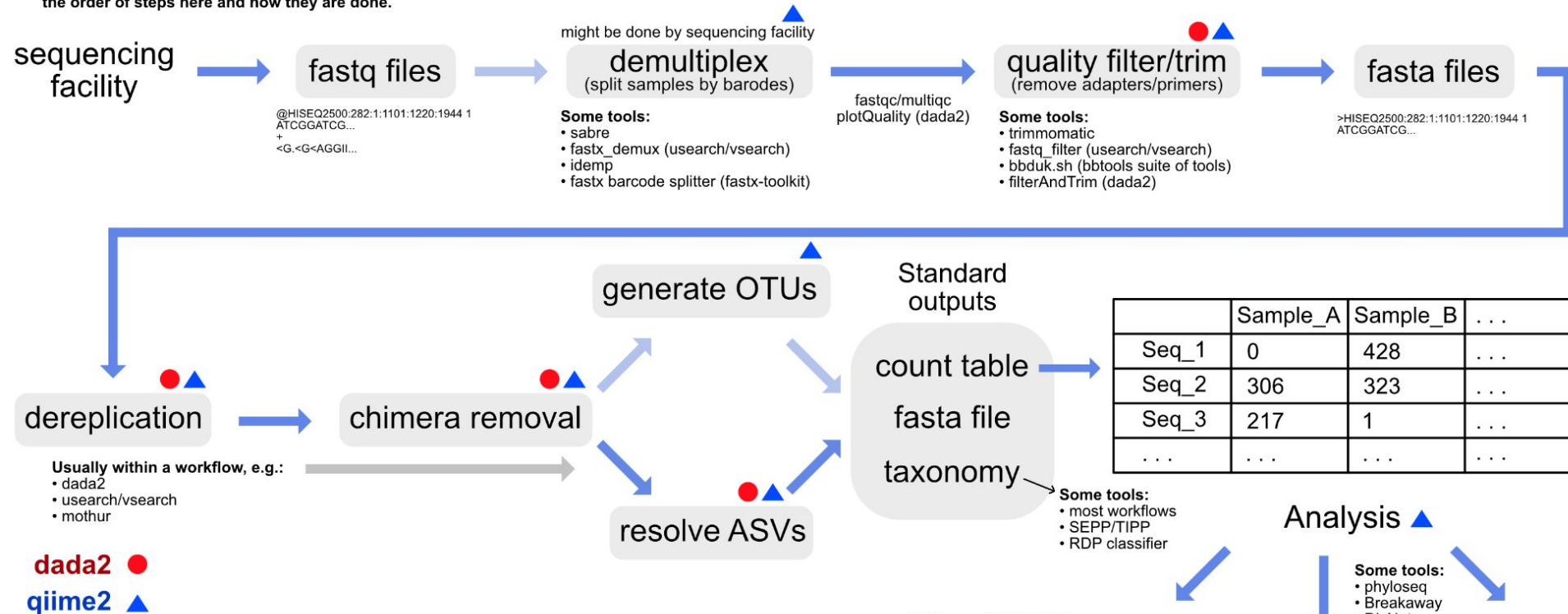
<https://unite.ut.ee/index.php>

Self-defined reference
sequences according to the
composition of community

Overview of generic* amplicon workflow

*This is generic; specific workflows can vary on the order of steps here and how they are done.

When working with your own data you should never follow any pipeline blindly. There can be critical differences based on your data.



Some tools that provide whole workflows:

dada2 runs within R (ASVs)

usearch/vsearch runs at the command line (ASVs and OTUs)

mothur runs at the command line (OTUs only currently)

qiime2 provides a multi-interface environment that employs processing tools like those above, infrastructure for easily documenting all processing performed, and interactive visualizations

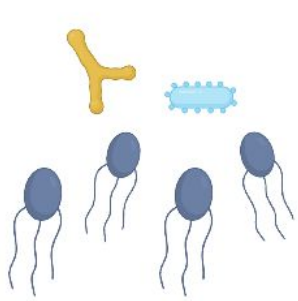
Diversity analysis of microbial community

- In (microbial) ecology, alpha diversity is the mean species diversity in sites or habitats at a local scale. The term was introduced by R. H. Whittaker (1972)
- Alpha diversity indices:
 - Species richness / Chao1
 - Shannon index
 - Simpson index
- Richness (number of microbial species) and evenness (similarity of each species' abundance)

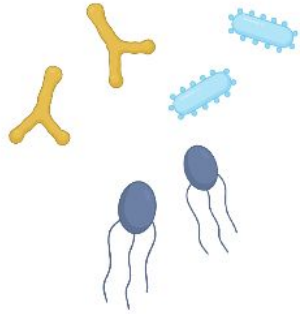
Diversity analysis of microbial community

- Beta diversity: ecological differences between samples
- Commonly used distance or dissimilarity metrics:
 - Bray-Curtis dissimilarity - quantitative and takes into consideration abundance and presence / absence
 - (Unweighted) Jaccard - qualitative, presence / absence
 - Weighted UniFrac - quantitative, similar to Bray-Curtis but takes into consideration phylogenetic relationships
 - Unweighted UniFrac - qualitative, like Jaccard focuses on presence / absence of taxa but also includes phylogenetic relationships

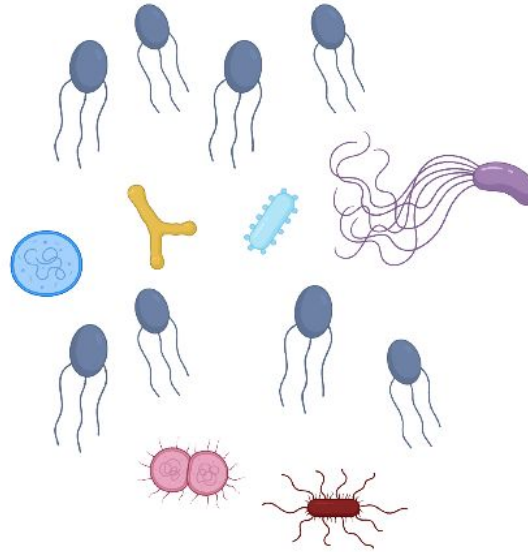
Example of alpha diversity of microbial community



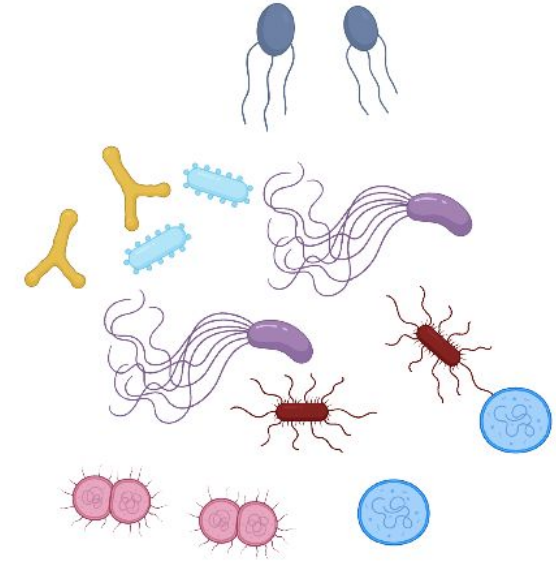
Low richness
Low evenness



Low richness
High evenness



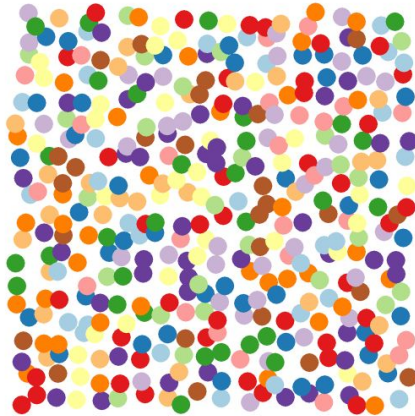
High richness
Low evenness



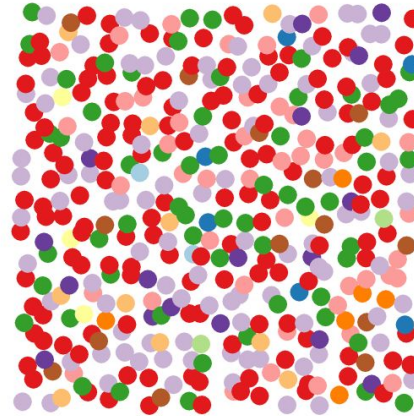
High richness
High evenness

Example of alpha diversity of microbial community

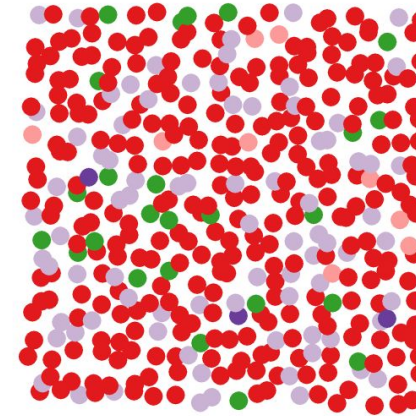
Community A
(perfectly even)



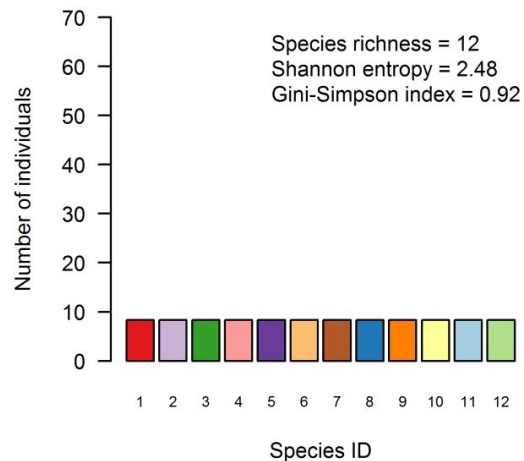
Community B
(moderately uneven)



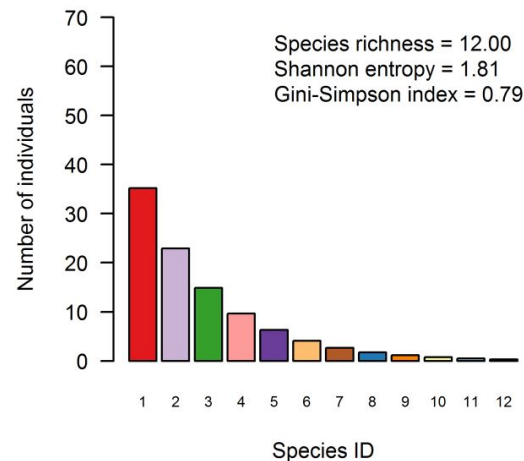
Community C
(highly uneven)



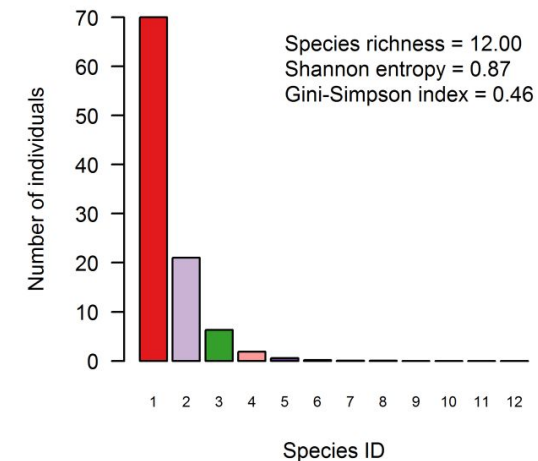
Species abundance distribution



Species abundance distribution



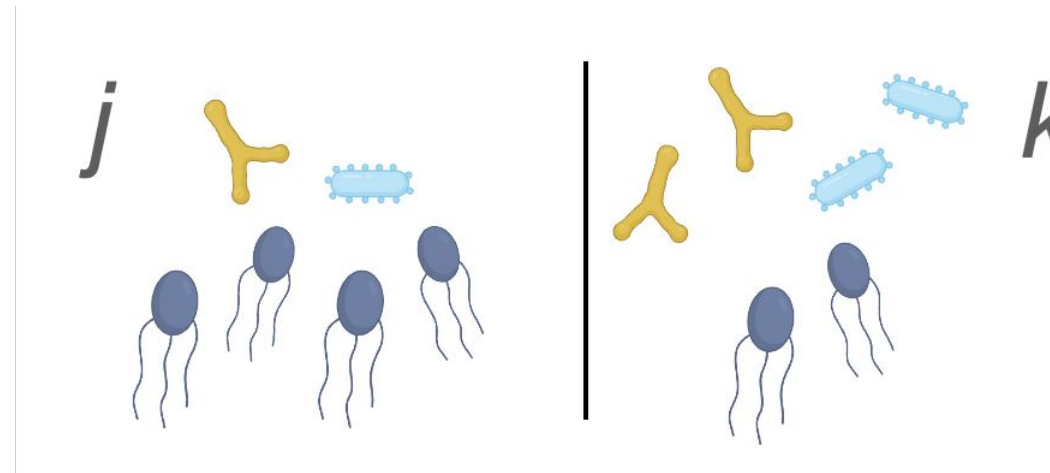
Species abundance distribution



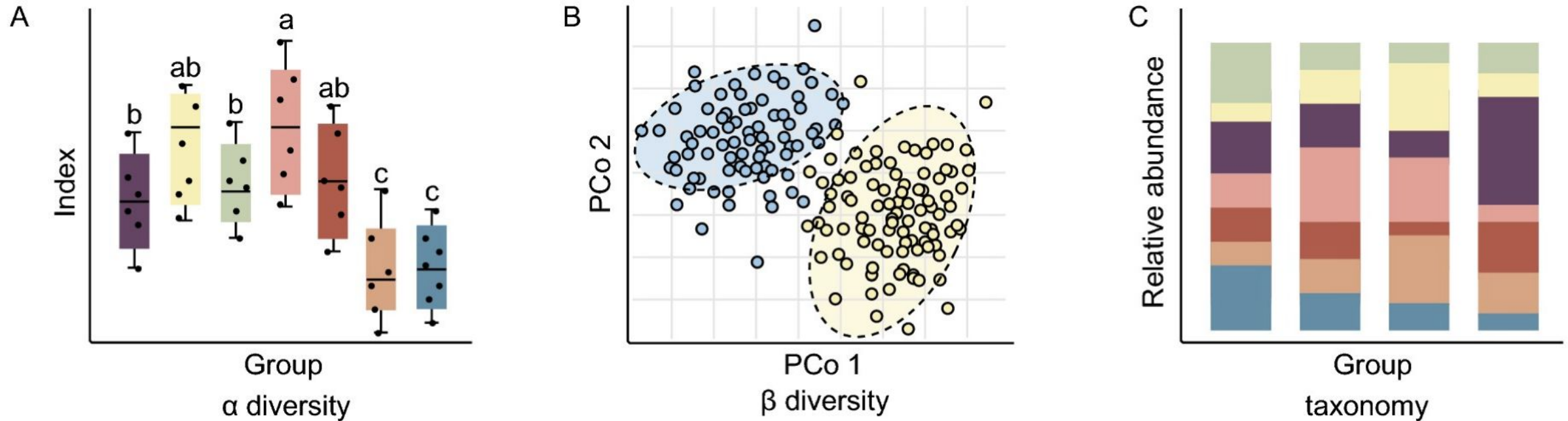
Example of beta diversity of microbial community

Bray–Curtis dissimilarity is a statistic used to quantify the dissimilarity in species composition between two different sites, based on counts at each site. It is named after J. Roger Bray and John T. Curtis who first presented it in a paper in 1957.

$$BC_{jk} = 1 - \frac{2C_{jk}}{S_j + S_k} = 1 - \frac{2 \sum_{i=1}^p \min(N_{ij}, N_{ik})}{\sum_{i=1}^p (N_{ij} + N_{ik})}$$



Example of downstream analysis of microbial community



INVITED REVIEW |  Open Access |   

Critical review of 16S rRNA gene sequencing workflow in microbiome studies: From primer selection to advanced data analysis

Alba Regueira-Iglesias, Carlos Balsa-Castro, Triana Blanco-Pintos, Inmaculada Tomás 

First published: 07 October 2023 | <https://doi.org/10.1111/omi.12434>

1

Clinical phase



- Selection of study groups
- Collection of oral samples
- Creation of a metadata table

2

Laboratory phase

- Bacterial DNA extraction
- PCR amplification of the 16S rRNA gene
 - Primer pair selection
- 16S rRNA gene sequencing
 - 1ST, 2ND, and 3RD generation technologies
 - Considerations for sequencing technology selection



3

Bioinformatics pipeline



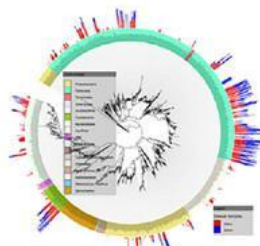
USEARCH
DADA²

Main
bioinformatic
pipelines

Pre-processing & quality control of raw sequence data



Taxonomy assignment

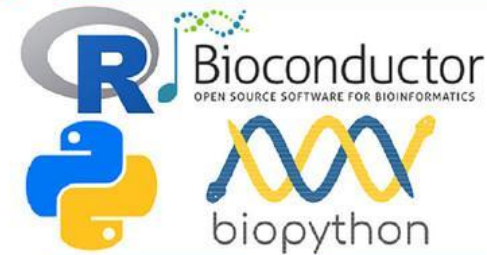


	ASVid	Sam1	Sam2	Sam3
1	ASVid	Sam1	Sam2	Sam3
2	AV00001	47024	75200	10863
3	AV00002	8911	2222	238
4	AV00003	12171	4857	32894
5	AV00004	3167	7808	2002
6	AV00005	17673	15341	501
7	AV00006	1972	6486	448
8	AV00007	348	865	32760

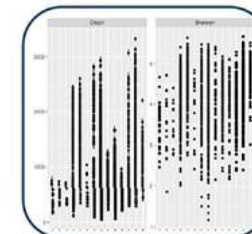
Phylogenetic tree &
abundance table of
OTUs/ASVs

4

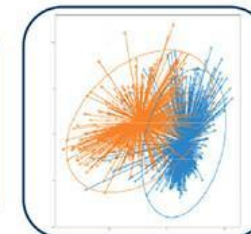
Advanced data analysis & visualisation



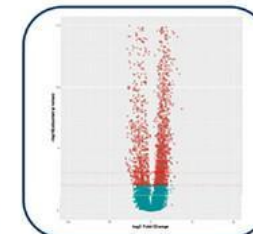
Analysis
packages



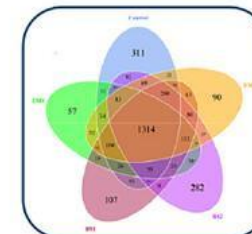
Alpha-diversity



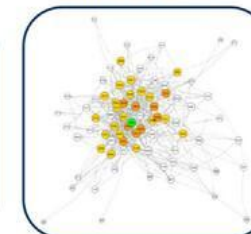
Beta-diversity
multivariate



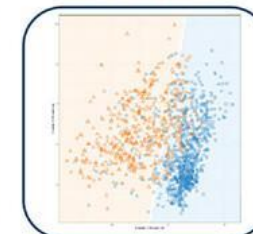
Beta-diversity
univariate



Core microbiome



Co-occurrence networks



Predictive modelling

Thanks for your attention!

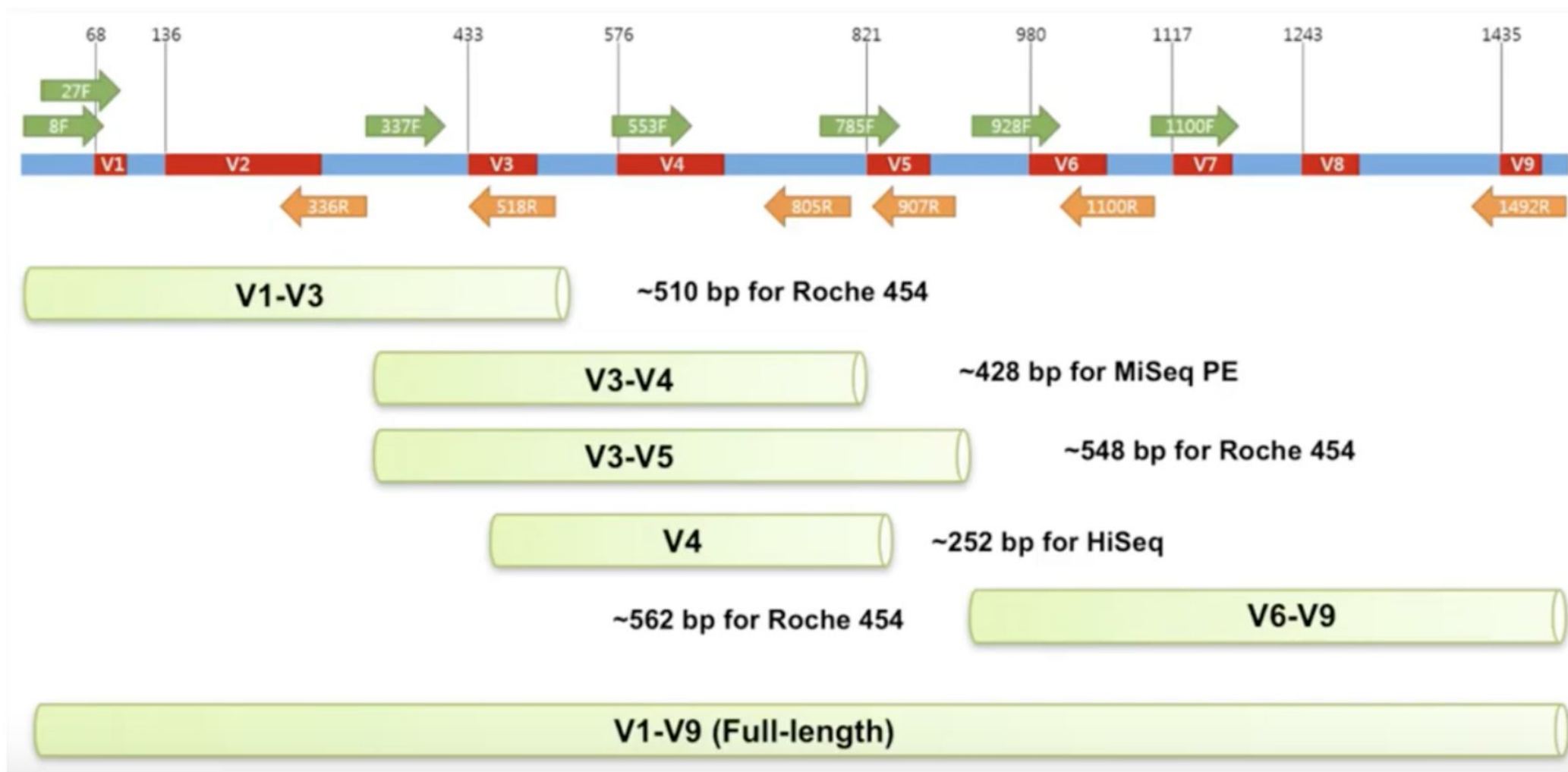


<https://www.menti.com/altarar89f96>

Join at menti.com | use code
3482 2634

Supplementary

Anatomy of a gene: 16S rDNA



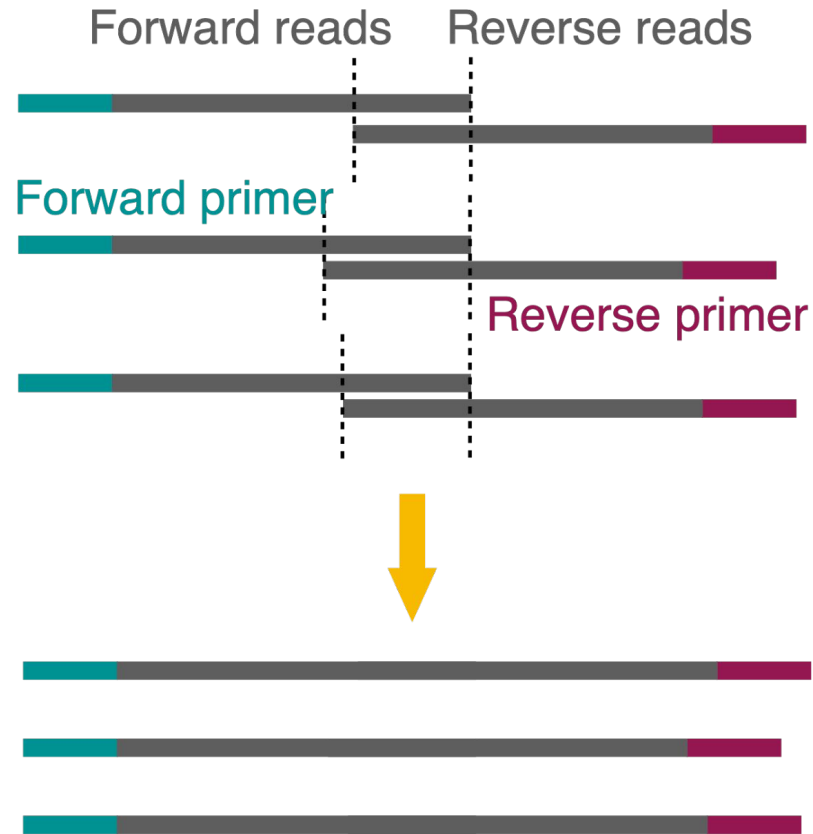
Qiime 2



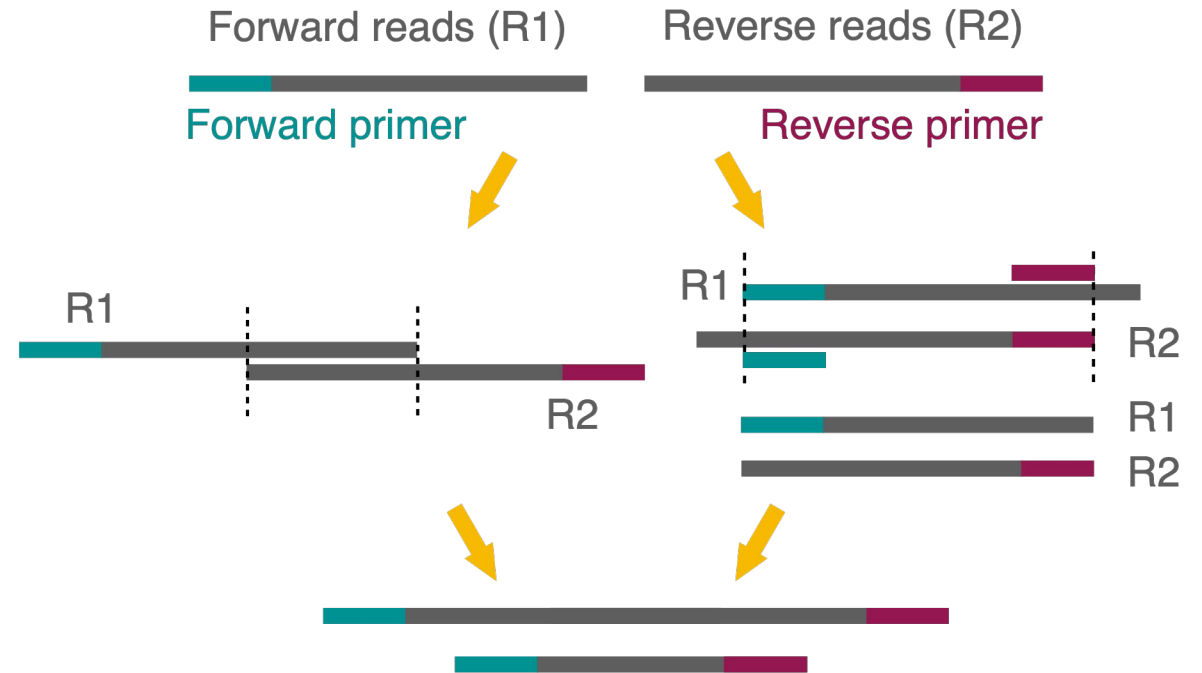
- a complete rewrite of Qiime 1; introduce the new concept of **artifacts** (input/output packages with metadata)
- strongly advocates **DADA2** algorithm for **ASV** detection rather than OTU picking
- new release are produced frequently, unfortunately sometimes changing some commands (not a sign of a mature and stable package)
- Still producing charts, in my opinion sometimes worse than in the past
- Robust **traceability** and reproducibility at the expense of ease of use and interchange of data with other tools

www.qiime2.org

Amplicon sequencing reads merge of bacteria and fungi



Pair-end reads merge of bacterial 16S rRNA gene sequencing.



Pair-end reads merge of fungal ITS gene sequencing.

Commonly used ordination methods in beta diversity

PCoA

- most common

- similar to PCA but works on distance metrics beyond euclidean

- maximizes linear correlation

- prone to the horseshoe effect (also observed in PCA)

UMAP (Uniform Manifold Approximation and Projection)

- non-linear

- can be used on multiple distance / dissimilarity metrics

- improved resolution in clusters

NMDS

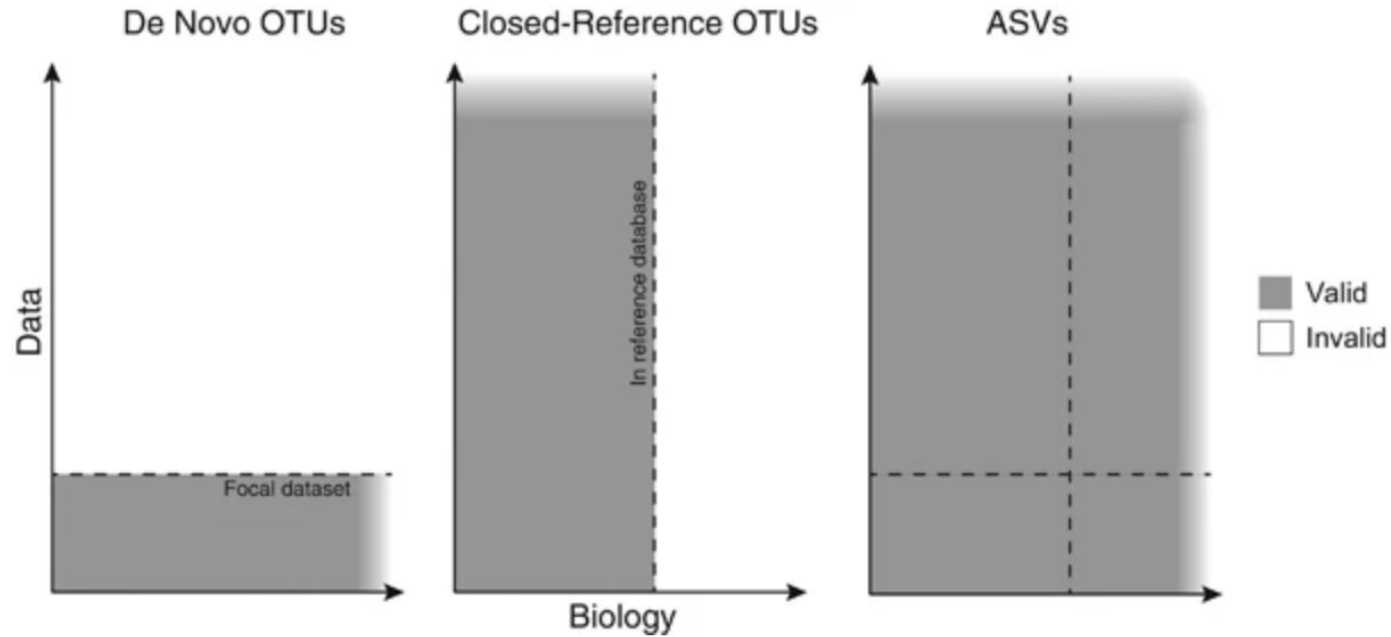
- better for rank ordered data (e.g., Bray-Curtis)

- dimensions are specified

- stress indicates how well the ordination represents the data (stress < 0.1 ~ good)

- no single solution

OTU clustering vs ASV



The extent of the validity of *de novo* OTUs, closed-reference OTUs and ASVs determined from a focal data set.

Pooling or not (sneak peek under the threshold)

- No pooling, full pooling, pseudo pooling
- We have to use relatively strict filtering options due to the high error rate of PCR based methods
- We exclude a lot of “low abundance” variants
- But maybe we are interested in these low abundance variants

Pooling or not (sneak peek under the threshold)

